The RSOPT Project

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Part I

Introduction

Chapter 1

About RSOPT

TODO

Chapter 2

Convex Optimization

TODO

Chapter 3

Software Concept

TODO

Part II

Convex Programs

Chapter 4

Linear Programs

TODO

Chapter 5

Quadratic Programs

This chapter covers various forms of **quadratic programs** supported by the RSOPT project. This document defines quadratic programs as convex programs whose objective has a constant Hessian with no negative eigenvalues.

5.1 Unconstrained Quadratic Programs

Unconstrained quadratic programs (`UnconstrainedQuadraticProgram`) are convex programs of the form

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top A x + b^\top x \tag{5.1}$$

where $A \in \mathbb{R}^{n \times n}$, $A \succeq 0$, and $b \in \mathbb{R}^n$. The relevant stationarity condition is

$$Ax^\star + b = 0 \Leftrightarrow x^\star = -A^{-1}b. \tag{5.2}$$

We solve unconstrained quadratic programs using the **conjugate gradient** (CG) method.

Definition. Two vectors p_i and p_j are *conjugate* with respect to A if $p_i^\top A p_j = 0$.

CG solves an n-dimensional system in at most n iterations by selecting conjugate search directions. The gradient of the objective at iterate x_k is

$$\nabla J_k = Ax_k + b. \tag{5.3}$$

CG selects $x_0 := 0$ and $p_0 := \nabla J_0 = b$ as the first search direction. Subsequent search directions p_k ($k > 0$) must be conjugate with all preceding search directions with respect to A. We write the recurrence as

$$p_{k+1} = Ax_{k+1} + b + \sum_{i=0}^{k} \beta_i p_i \tag{5.4}$$

and constrain β_i using the definition of conjugate vectors. For all $j = 0, 1, \cdots k$, we have

$$(Ax_{k+1} + b + \sum_{i=0}^{k} \beta_i p_i)^\top A p_j = 0. \tag{5.5}$$

Expanding gives us

$$x_{k+1}^\top A^\top A p_j + b^\top A p_j + \sum_{i=0}^{k} \beta_i p_i^\top A p_j = 0. \tag{5.6}$$

Notice that, by our "inductive hypothesis", $p_0, p_1, \cdots p_{k-1}$ are already pairwise conjugate with respect to A. We can now focus only on the case $j = k$ and take all other $\beta_i = 0$.

$$x_{k+1}^\top A^\top A p_k + b^\top A p_k + \beta_k p_k^\top A p_k = 0. \tag{5.7}$$

This gives us a closed form for β_k.

$$\beta_k = -\frac{(\nabla J_{k+1})^\top A p_k}{p_k^\top A p_k} \tag{5.8}$$

Therefore, the step direction p_k for $k > 0$ is given by

$$\boxed{p_{k+1} = \nabla J_{k+1} - \frac{(\nabla J_{k+1})^\top A p_k}{p_k^\top A p_k} p_k.} \tag{5.9}$$

After selecting the step direction, it is necessary to select the step size α_k. We select α_k according to the QP

$$\min_{\alpha_k \in \mathbb{R}} \frac{1}{2}(x_k + \alpha_k p_k)^\top A (x_k + \alpha_k p_k) + b^\top (x_k + \alpha_k p_k) \tag{5.10}$$

Expanding the objective and dropping all terms not dependent on α, we have

$$\frac{d}{d\alpha_k}\left[\alpha_k p_k^\top A x_k + \frac{1}{2}\alpha_k^2 p_k^\top A p_k + \alpha_k b^\top p_k\right] = 0. \tag{5.11}$$

We find that α_k is also determined analytically as

$$\boxed{\alpha_k = \frac{-(\nabla J_k)^\top p_k}{p_k^\top A p_k}.} \tag{5.12}$$

The complete CG algorithm is detailed in Algorithm 1.

Algorithm 1 Conjugate Gradient

Require: A, b

1: $p_0 \leftarrow b$
2: **while do**
3: **end while**

5.2 Equality-Constrained Quadratic Programs

TODO

5.3 Inequality-Constrained Quadratic Programs

TODO